Investing in the green transition and competition from laggards

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The views expressed here are not representative of the Bank of England's.

Motivation

- green transition, i.e. the transformation of the economy towards climate neutrality, one of major challenges
- green transition requires firms to invest in the reduction of carbon emissions originating in the production process (= *greening investment*)
- during the transition, investing and non-investing firms compete in the same market

Which trade-offs exist for the policy-maker when incentivising a (slower/faster) green transition?

Results:

- trade-off btw. output stabilisation and the share of investors
- policy-maker can leverage a competition effect
- competition by laggards hinders the green transition

Outline

Greening investment facts

Small model

Outlook: full model

Literature

green transition

- Fried (2018); Hassler et al. (2021); Golosov et al. (2014) (carbon taxes) $\Rightarrow$ we add heterogeneous firms

environmental heterogeneous firms

- Anouliès (2017); Dardati (2016); Dardati and Saygili (2020); Konishi and Tarui (2015); Qiu et al. (2018) $\Rightarrow$ we add competition effects

dynamics of heterogenous agents

- Boppart et al. (2018) $\Rightarrow$ we investigate the green transition

environmental regulation & competition

- Jondeau et al. (2023): IAM with abatement sector; subsidies affect competition and entry into abatement sector

- **Greening investment facts**

Data

- data from Kreditanstalt für Wiederaufbau (KfW-Bank)
- cross-sectional survey data
- 12,784 German firms
- only SMEs (turnover $\leq$ 500 million Euro)
- years 2021 & 2022

Climate investment

Survey definition **climate investment**[1]:

"Climate protection investments are **investments in measures to avoid or reduce greenhouse gas emissions** *in your company. These include investments to save energy or increase energy efficiency, measures to utilise renewable energies or investments in climate-friendly transport, such as the purchase of electric vehicles."*

⇒ a proxy for *greening investment*

[1] Translation, original in German.

Fact 1: Productivity and share of green investors (+)

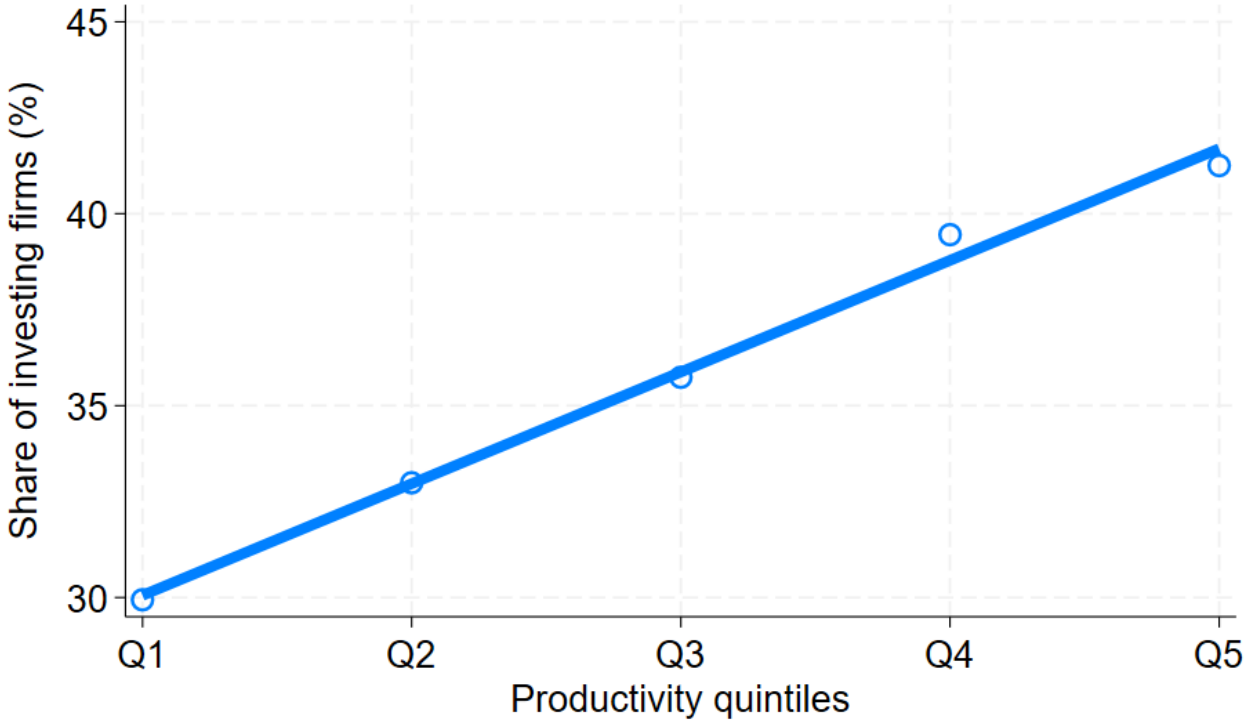


Figure: Relation between productivity and greening investment

$\Rightarrow$ increase in productivity (turnover/employees) by 1% associated to increase in likelihood of green investing of 3-4%

Fact 2: Firm dynamics and investment intensity (+)

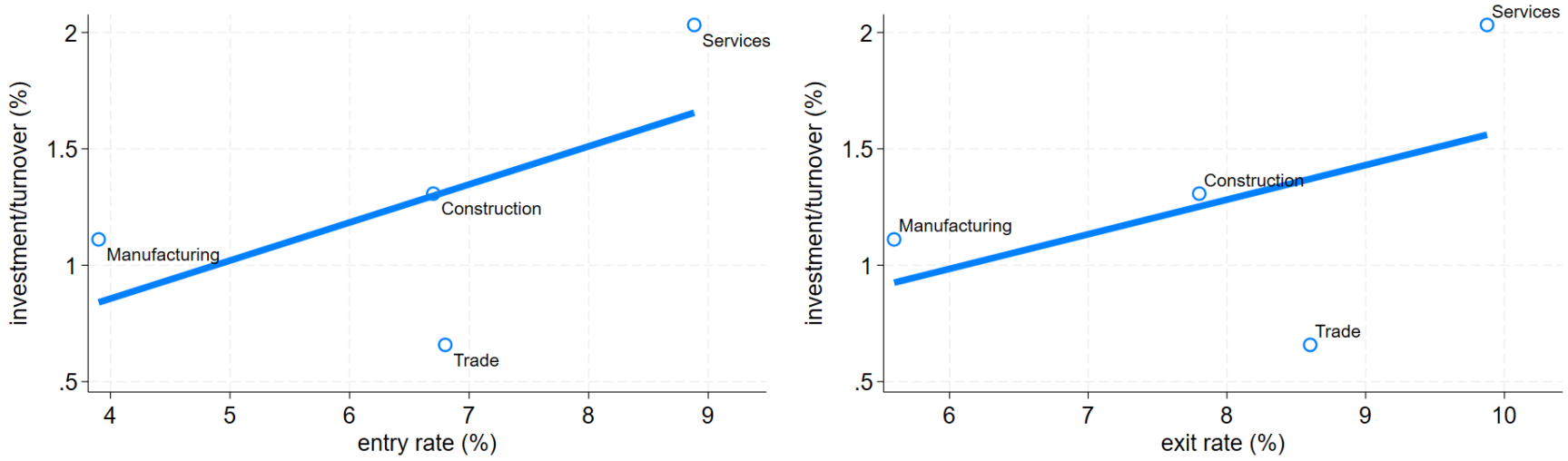


Figure: Sectoral entry rates and exit rates relative to investment intensity (only 2021, N=1153)

⇒ our argument: firms in less dynamic sectors invest less (e.g. manufacturing) due to more competition from incumbent non-investing firms

Fact 2: Firm dynamics and investment intensity (+)

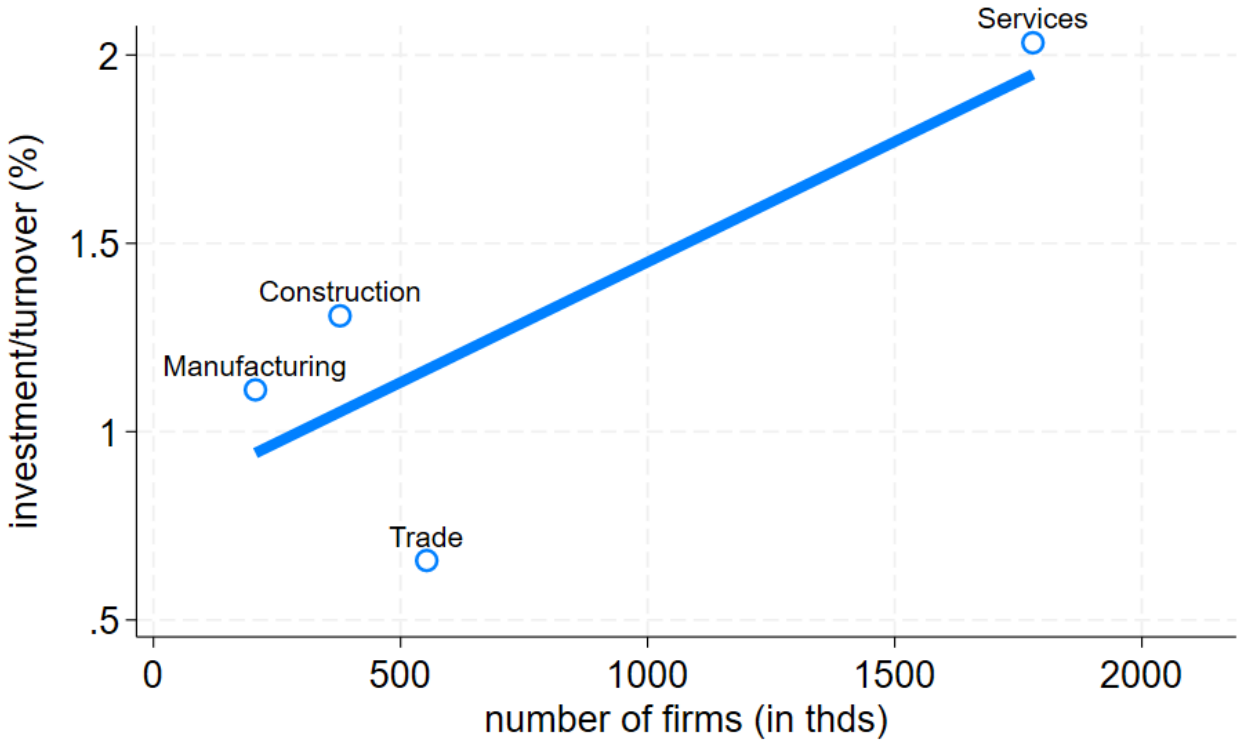


Figure: Sector size relative to investment intensity (only 2021, N=1153)

- **Small model**

Setup

Period 1:

- productivity draw $z_1 \Rightarrow$ if firm value $V < 0$ firms exit period 2
- firms produce & get profits $\pi_1 = p_1 \ \exp(z_1) - f$
- firms make greening investments at cost $c > 0$ ($\partial c / \partial \pi < 0$), or not ($c = 0$) $\Rightarrow \tilde{z}_I$

Period 2:

- low-productivity firms ($z_1 < \tilde{z}_P$) exit
- new productivity draw, production, profits $\pi_2(z_2)$
- probability $\kappa > 0$: enactment of transition policy that mandates all firms to use a green production process
  - $c > 0$: firms produce
  - $c = 0$: firms do not produce

Greening investment decision

Firms invest if:

$$V_I > V_{NI} \tag{1}$$

$$\pi_1(z_1, p_1) + \kappa\ E[\pi_2(z_2, p_2)] + (1-\kappa)\ E[\pi_2(z_2, p_2)] - c(\pi_1(z_1, p_1))$$
$$> \pi_1(z_1, p_1) + (1-\kappa)\ E[\pi_2(z_2, p_2)] \tag{2}$$

$$\kappa\ E[\pi_2(z_2, p_2)] > c(\pi_1(z_1, p_1)) \tag{3}$$

$\Rightarrow$ true to Fact 1:

$z_2 \uparrow \quad LHS \uparrow$

$z_1 \uparrow \quad \pi_1 \uparrow \quad c \downarrow \quad RHS \downarrow$

Investment decision margin

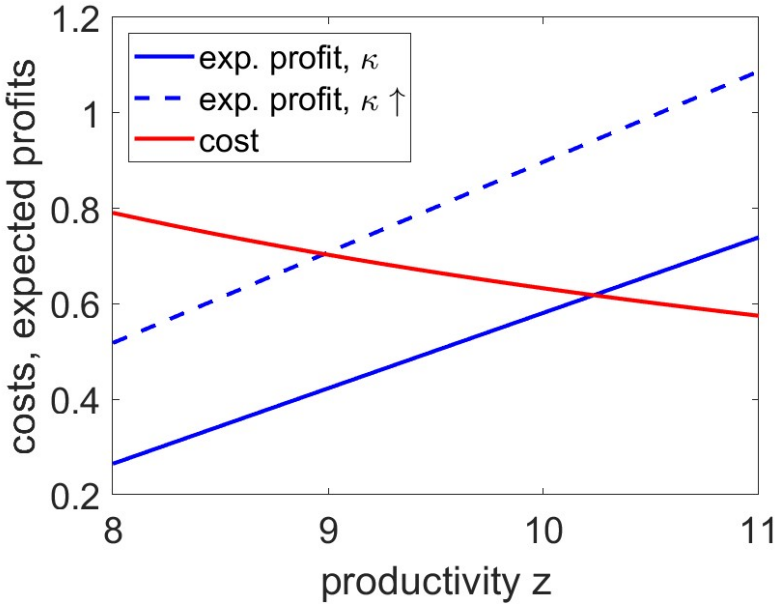


Figure: Investment decision in the small model

$$\kappa \; E[\pi_2(z_2, p_2)] > c(\pi_1(z_1, p_1))$$

$\Rightarrow$ an increase in the policy probability increases the share of investors

Optimal policy

$$\max_{Y_2(\kappa),\, Ni_1(\kappa)} \log E[Y_2(N_2(\kappa))] + \log\left(\frac{Ni_1(\tilde{z}_I(\kappa))}{N_1}\right)$$

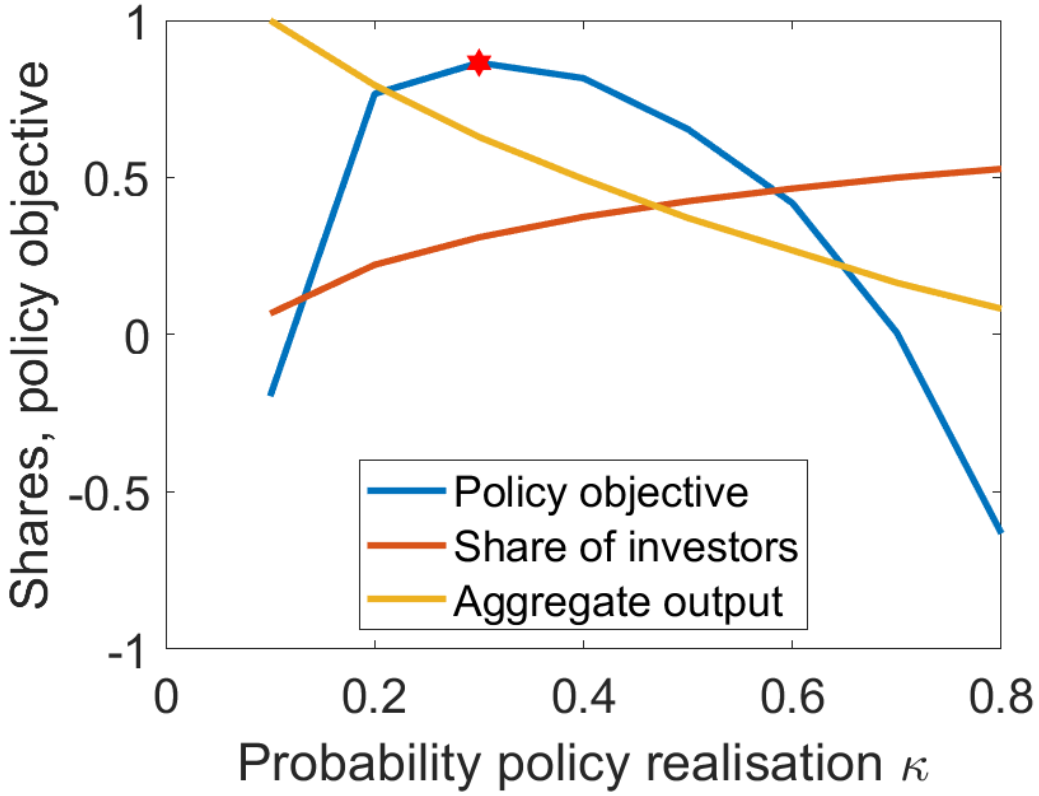
Investment channel:
$\kappa \uparrow \quad \tilde{z}_I \downarrow \quad Ni_1 \uparrow \quad Ni_1/N_1 \uparrow$

Output channel:
$\kappa \uparrow \quad \tilde{z}_P \uparrow \quad V_{NI} \downarrow \quad N_2 \downarrow \quad Y_2 \downarrow$

$V_{NI} = \pi_1(z_1, p_1) + (1-\kappa)\, E[\pi_2(z_2, p_2)]$: firm value non-investor
$\tilde{z}_I$: investment cutoff
$\tilde{z}_P$: production cutoff

assumption here: $p_1 = p_2 = p$ exogenous

Result 1: policy trade-off $E[Y_2]$ vs. Ni_1/N_1



Note: output relative max level; policy objective scaled for presentation purposes

$\Rightarrow$ there is a trade-off in setting an optimal κ between output stabilisation and incentivising a green transition

Competition effect

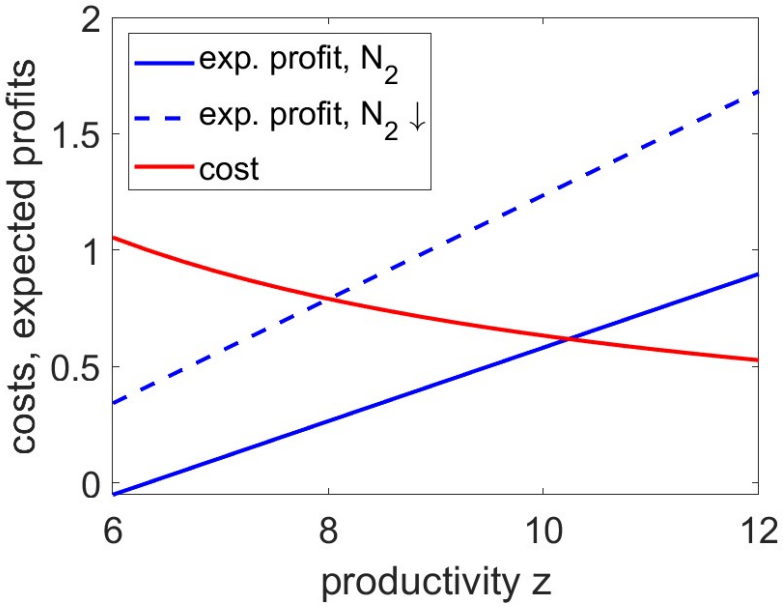
Key assumption: $\frac{\partial p}{\partial N} < 0$

- the current price level is a function of the number of firms
- more competition lowers the aggregate price level
- we assume Dixit-Stiglitz-price aggregation: $p = N^{\frac{1}{1-\sigma}}$

$$\Rightarrow \kappa\, E[\pi_2(z_2, \underbrace{p_2(N_2)}_{\textbf{effect 1}})] > c(\pi_1(z_1, \underbrace{p_1(N_1)}_{\textbf{effect 2}})$$

effect 1: can be leveraged by policy-maker
effect 2: competition by "laggards"

Result 2: competition effect 1

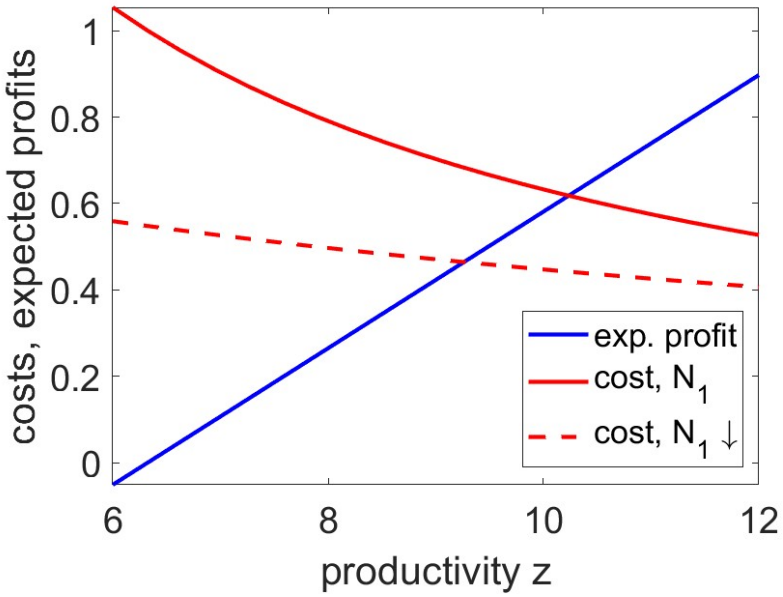


$$\kappa\ E[\pi_2(z_2, p_2(N_2))] > c(\pi_1(z_1, p_1(N_1)))$$

$$\kappa \uparrow \quad \tilde{z}_P \uparrow \quad V_{NI} \downarrow \quad N_2 \downarrow \quad \underline{p_2 \uparrow} \quad \underline{\pi_2 \uparrow \quad \tilde{z}_I \downarrow \quad Ni_1/N_1 \uparrow}$$

$\Rightarrow$ the policy-maker can leverage a competition effect induced by κ when incentivising a green transition

Result 3: competition effect 2



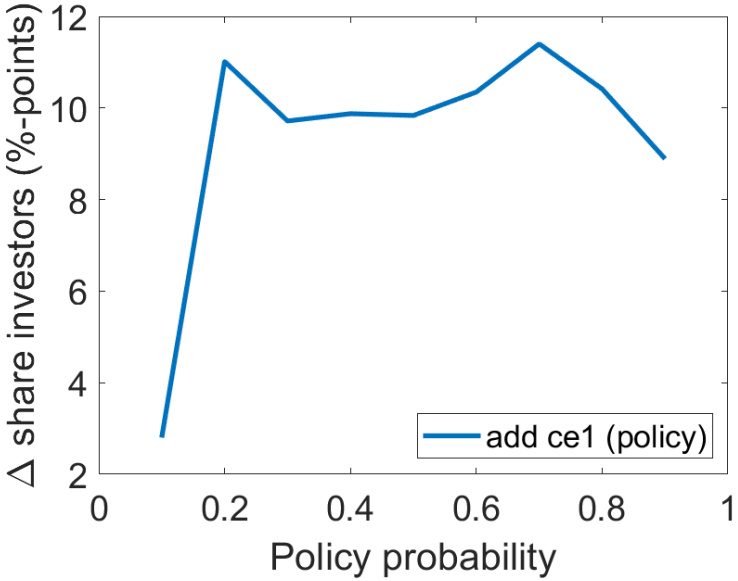
$$\kappa\ E[\pi_2(z_2, p_2(N_2))] > c(\pi_1(z_1, p_1(N_1)))$$

$N_1 \uparrow \quad p_1 \downarrow \quad \pi_1 \downarrow \quad c \uparrow \quad \tilde{z}_I \uparrow \quad Ni_1/N_1 \downarrow$

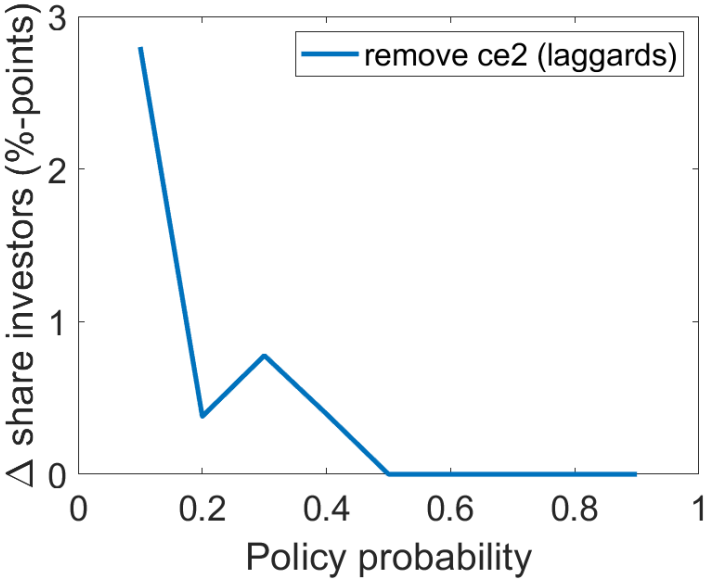
$\Rightarrow$ competition by incumbent non-investors ("laggards") can be an opposing force to a green transition

Transition experiment: Δ share of investors

Experiment: change of share of investors following sequential increase in policy probability κ



(a) Competition effect 1 relative to none



(b) Both competition effects relative to only ce1

$\Rightarrow$ competition effect leveraged by policy-maker (ce1) speeds up transition, whereas competition from "laggards" (ce2) hinders it

Outlook: full model

- extension of Hopenhayn (1992)
- binary one-sided one-time green investment decision
  - firms transition into the pool of investors with one-time cost
- policy experiments
  1. comparing stationary equilibria
  2. MIT-shock: sequential equilibrium path of green policy parameters $\{\kappa\}_{t=0}^{\infty}$
- on 2.: conditional on the speed of the policy introduction...
  - how long will there be long-standing incumbent non-investing firms in the market? how quickly can the green transition be achieved?
  - how does this differ across sectors (higher/lower entry costs $\rightarrow$ higher/lower entry rates $\rightarrow$ higher share of green entrants)?

Conclusion

- greening investment positively correlated with productivity in the data
- build heterogeneous firms model à la Hopenhayn (1992) where firms transition to green production with one-time one-sided investment decision
- there is a trade-off in setting an optimal κ between output stabilisation and incentivising a green transition
- the policy-maker can leverage a competition effect that κ induces when incentivising a green transition
- competition by incumbent firms ("laggards") can be an opposing force to a green transition

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Data: descriptives

	count	%
Manufacturing	3392	21.27
Construction	2506	15.71
Services	4275	26.80
Trade	5050	31.66
Other	726	4.55
Total	15949	100.00
N	15949	

Table: Sectoral breakdown

Data: descriptives

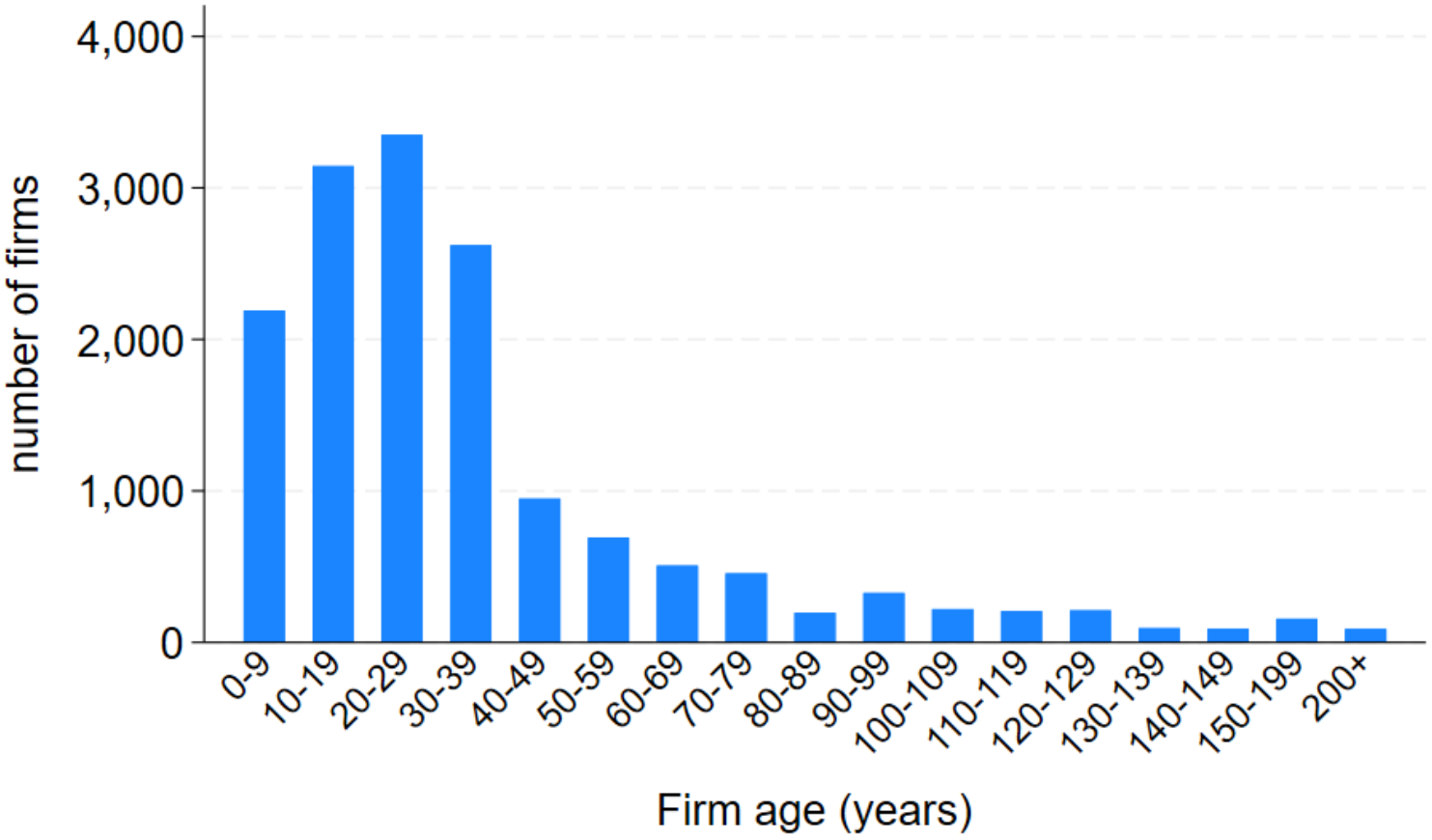


Figure: Data: age distribution

Data: descriptives

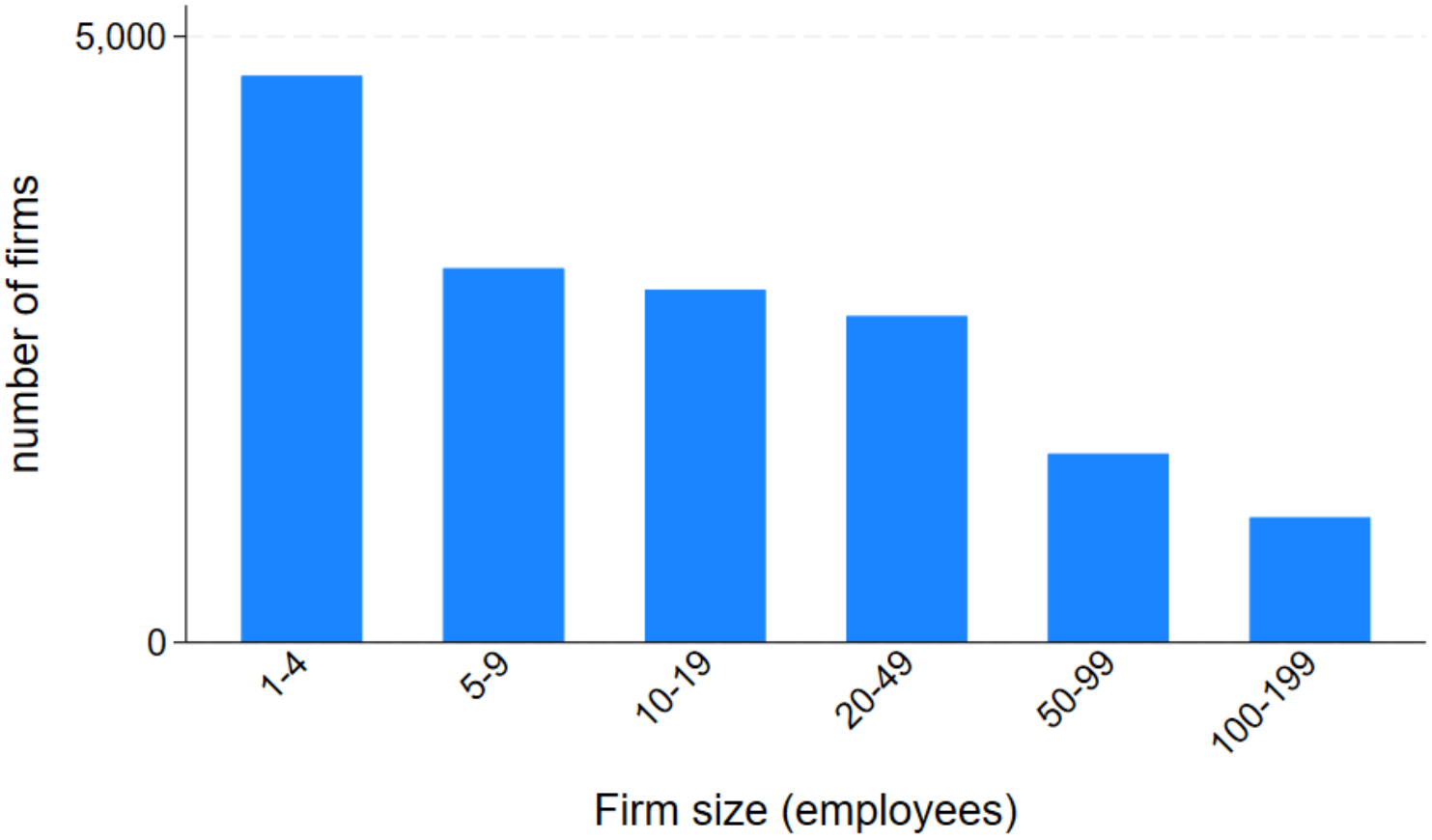


Figure: Data: size distribution (nr. of employees)

Share of green investors by sector

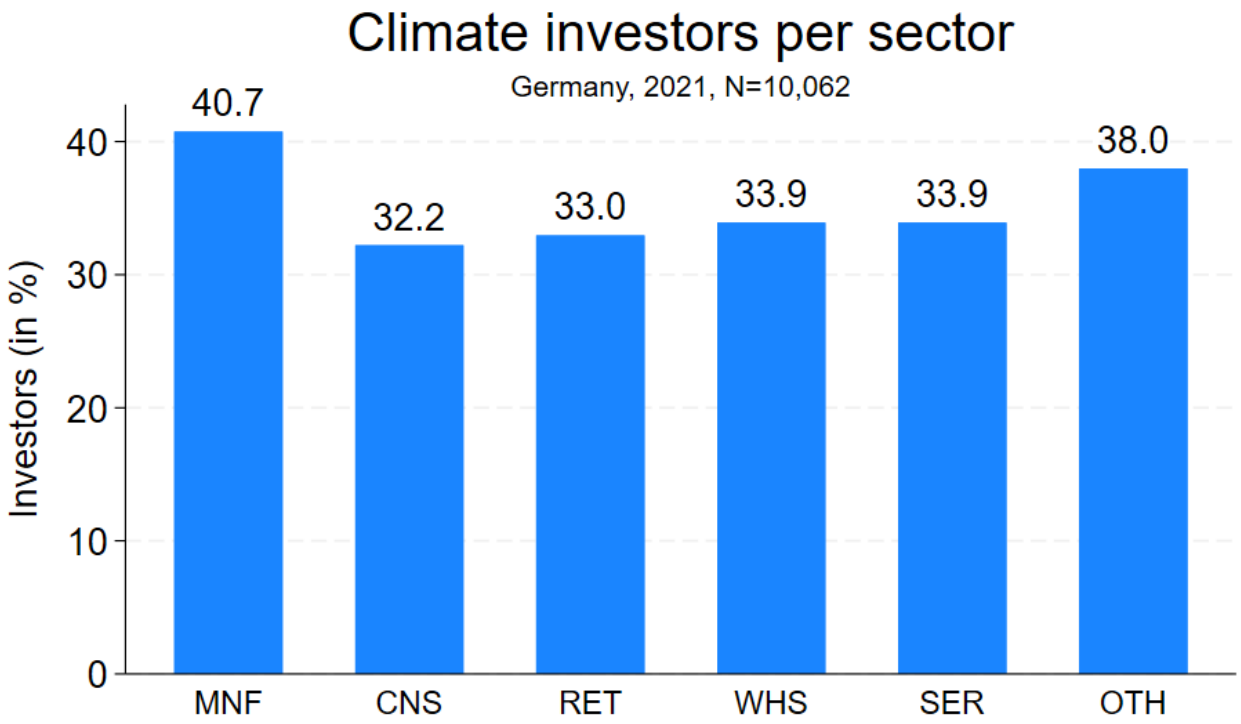


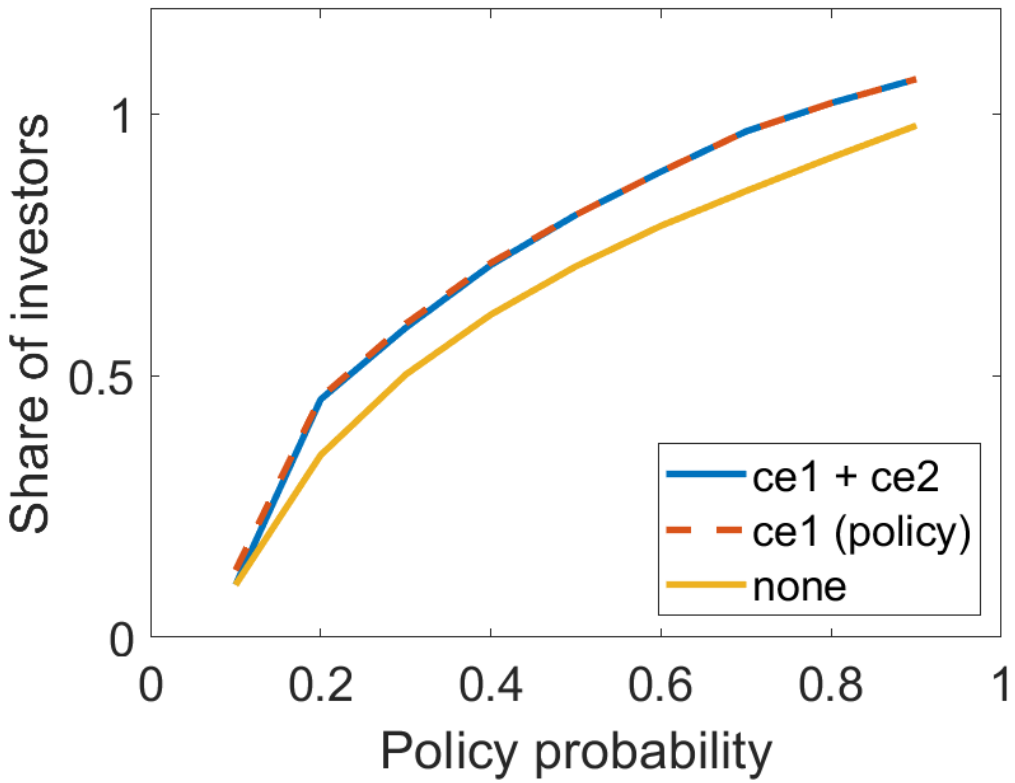
Figure: Share of green investors by sector

⇒ Manufacturing disposes of most investors

Appendix

	(1)	(2)	(3)	(4)	(5)
Prod (log)	0.0356***	0.0347***	0.0435***	0.0381***	0.0388***
Size	0.0009***			0.0009***	0.00099***
Age		0.001***			0.0009***
Manuf			0	0	0
Constr			-0.088***	-0.0658***	-0.0462**
Services			-0.0597***	-0.0564***	-0.0453**
Other			-0.0341	-0.0025	-0.009
Trade			-0.097***	-0.0712***	-0.053***
Energy					✓
Obs	12260	11962	12260	12260	9298

t statistics in parentheses; * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$; energy intensity relative to $< 2\%$

Table: Likelihood of climate investment (logit marginal effects)

Transition experiment: share of investors